Scope of Work and Preliminary EDA

Comprehensive Data Preprocessing for Machine Learning

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Project Statement

We have selected the project topic: **Sentiment Analysis in Tourism Platforms**. The goal of this project is to develop a comprehensive data preprocessing pipeline that prepares tourism-related text data (such as hotel reviews, travel forum posts, and destination comments) for sentiment classification. Our project will focus on:

- Collecting and integrating text data from various tourism platforms and review sites
- Performing extensive text cleaning, normalization, and preprocessing specific to tourism domain language
- Engineering features including sentiment scores, tourism-specific keywords, and linguistic patterns
- Building a reusable preprocessing pipeline for tourism-related NLP tasks
- Evaluating the impact of preprocessing on sentiment classification model performance

Preliminary EDA Plan

We plan to conduct an exploratory data analysis (EDA) to understand the structure, quality, and characteristics of our tourism sentiment dataset. The EDA will include the following steps:

1. Data Loading and Overview

- Load the dataset and examine its size, columns, and data types
- Check for missing values, duplicates, and inconsistencies in the data

- Analyze the distribution of sentiment labels (positive, negative, neutral) and rating scores
- Examine basic statistics of the dataset

2. Text-Specific Exploration

- Analyze text length distribution (word count, character count) across different sentiment classes
- Identify common words, phrases, and tourism-specific terminology using frequency analysis
- Explore the relationship between review ratings and text sentiment
- Examine patterns in review timing, platform sources, and user demographics if available

3. Visualization

- Generate word clouds for each sentiment class to visualize frequently used terms
- Create bar charts for top N-grams (unigrams, bigrams) in positive vs negative reviews
- Plot distribution of review lengths across different sentiment categories
- Visualize rating distributions and their correlation with sentiment labels
- Create time series plots if temporal data is available

4. Initial Insights and Hypothesis

- Formulate hypotheses about which features may be most predictive of sentiment
- Identify potential challenges such as class imbalance, sarcasm, or domain-specific language
- Note any data quality issues that need to be addressed in preprocessing

5. Tools and Libraries

We will use the following Python libraries for EDA:

- `pandas`, `numpy` for data manipulation
- `matplotlib`, `seaborn`, `plotly` for visualization
- `nltk`, `spaCy`, `wordcloud` for text analysis
- `textblob`, `vaderSentiment` for preliminary sentiment analysis

Next Steps

After completing the EDA, we will:

- Refine our project question based on EDA insights
- Begin data cleaning and feature engineering specific to tourism domain
- Develop a comprehensive preprocessing pipeline for text data
- Train and evaluate baseline sentiment classification models
- Compare model performance before and after preprocessing